ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 144°	One part $= 360^\circ \div 15 = 24^\circ$.
2	**(d)** 7	Opposite angles are equal.
3	**(c)** 20°	Adjacent angles add up to 180°.
4	**(a)** 28 cm	Diagonals bisect each other, so $AC = 2 \times OA$.
5	**(d)** parallelogram	A test for a parallelogram.
6	**(c)** 25°	A diagonal of a rhombus bisects the angle at the vertex.
7	**(a)** 43 cm	Each new side is half of a side of the original triangle.
8	**(c)** 32 cm^2	Area $= \frac{1}{2}d^2$.
9	**(d)** 180°	One part $= 360^\circ \div 12 = 30^\circ$.
10	**(b)** 29	Opposite angles are equal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle C = \angle A = 88^\circ$ (opposite angles); $\angle B = 180^\circ - 88^\circ = 92^\circ$ (adjacent angles) (1); $\angle D = \angle B = 92^\circ$ (1).	1+1
12	Diagonal $= \sqrt{8^2 + 15^2}$; $= \sqrt{289} = 17$ cm (1); Perimeter $= 2(8 + 15) = 46$ cm (1).	1+1
13	$EF = 17$ cm, $FD = 25$ cm, $DE = 12$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 17 + 25 + 12 = 54$ cm (1).	1+1
14	$\angle C = \angle A = 40^\circ$ (opposite angles); $\angle B = 180^\circ - 40^\circ = 140^\circ$ (adjacent angles) (1); $\angle D = \angle B = 140^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Opposite angles are equal; $2x + 8 = x + 16$ gives $x = 8$ (1); Angles $A = C = 24^\circ$ (1); $B = D = 180^\circ - 24^\circ = 156^\circ$ (1).	1+1+1
16	Adjacent angles add up to 180°: $x + 8x = 180^\circ$ (1); $x = 20^\circ$ and the other is 160° (1); Opposite angles are equal: the four angles are $20^\circ, 160^\circ, 20^\circ, 160^\circ$ (1).	1+1+1
17	Diagonals bisect each other, so $OA = OC$; $4x + 2 = 3x + 4$ gives $x = 2$ (1); $OA = 10$ cm (1); $AC = 2 \times 10 = 20$ cm (1).	1+1+1
	OR: Opposite angles are equal; $2x + 2 = x + 15$ gives $x = 13$ (1); Angles $A = C = 28^\circ$ (1); $B = D = 180^\circ - 28^\circ = 152^\circ$ (1).	1+1+1
18	Adjacent angles add up to 180°: $x + 9x = 180^\circ$ (1); $x = 18^\circ$ and the other is 162° (1); Opposite angles are equal: the four angles are $18^\circ, 162^\circ, 18^\circ, 162^\circ$ (1).	1+1+1
	OR: Diagonals bisect each other, so $OA = OC$; $5x + 3 = 3x + 13$ gives $x = 5$ (1); $OA = 28$ cm (1); $AC = 2 \times 28 = 56$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AD = 14$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 19$ cm (1); Hence $AE = EC$ (2).	1+1+1+2
	OR: In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 8.5$ cm and $QR = PS = 7.5$ cm (1); Perimeter $= 17 + 15 = 32$ cm (2).	1+1+1+2
20	$AD = 13$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 16$ cm (1); Hence $AE = EC$ (2).	1+1+1+2
	OR: In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 6.5$ cm and $QR = PS = 8$ cm (1); Perimeter $= 13 + 16 = 29$ cm (2).	1+1+1+2

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 30°	Adjacent angles add up to 180°.
2	**(d)** 30 cm	Diagonals bisect each other, so $AC = 2 \times OA$.
3	**(b)** parallel to the third side and half of it	Midpoint theorem.
4	**(a)** 57°	A diagonal of a rhombus bisects the angle at the vertex.
5	**(b)** 48 cm	Each new side is half of a side of the original triangle.
6	**(a)** 50 cm^2	Area $= \frac{1}{2}d^2$.
7	**(c)** 144°	One part $= 360^\circ \div 15 = 24^\circ$.
8	**(a)** 24	Opposite angles are equal.
9	**(b)** 45°	Adjacent angles add up to 180°.
10	**(d)** 28 cm	Diagonals bisect each other, so $AC = 2 \times OA$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diagonal $= \sqrt{10^2 + 24^2}$; $= \sqrt{676} = 26$ cm (1); Perimeter $= 2(10 + 24) = 68$ cm (1).	1+1
12	$EF = 12$ cm, $FD = 9$ cm, $DE = 17$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 12 + 9 + 17 = 38$ cm (1).	1+1
13	$\angle C = \angle A = 126^\circ$ (opposite angles); $\angle B = 180^\circ - 126^\circ = 54^\circ$ (adjacent angles) (1); $\angle D = \angle B = 54^\circ$ (1).	1+1
14	Diagonal $= \sqrt{16^2 + 30^2}$; $= \sqrt{1156} = 34$ cm (1); Perimeter $= 2(16 + 30) = 92$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Opposite angles are equal; $4x + 10 = x + 61$ gives $x = 17$ (1); Angles $A = C = 78^\circ$ (1); $B = D = 180^\circ - 78^\circ = 102^\circ$ (1).	1+1+1
16	Adjacent angles add up to 180°: $x + 4x = 180^\circ$ (1); $x = 36^\circ$ and the other is 144° (1); Opposite angles are equal: the four angles are $36^\circ, 144^\circ, 36^\circ, 144^\circ$ (1).	1+1+1
17	Diagonals bisect each other, so $OA = OC$; $4x + 4 = x + 31$ gives $x = 9$ (1); $OA = 40$ cm (1); $AC = 2 \times 40 = 80$ cm (1).	1+1+1
	OR: Opposite angles are equal; $3x + 4 = x + 30$ gives $x = 13$ (1); Angles $A = C = 43^\circ$ (1); $B = D = 180^\circ - 43^\circ = 137^\circ$ (1).	1+1+1
18	Adjacent angles add up to 180°: $x + 5x = 180^\circ$ (1); $x = 30^\circ$ and the other is 150° (1); Opposite angles are equal: the four angles are $30^\circ, 150^\circ, 30^\circ, 150^\circ$ (1).	1+1+1
	OR: Diagonals bisect each other, so $OA = OC$; $4x + 1 = 2x + 9$ gives $x = 4$ (1); $OA = 17$ cm (1); $AC = 2 \times 17 = 34$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AD = 23$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 17$ cm (1); Hence $AE = EC$ (2).	1+1+1+2
	OR: In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 15$ cm and $QR = PS = 7.5$ cm (1); Perimeter $= 30 + 15 = 45$ cm (2).	1+1+1+2
20	$AD = 8$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 13$ cm (1); Hence $AE = EC$ (2).	1+1+1+2
	OR: In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 5.5$ cm and $QR = PS = 9$ cm (1); Perimeter $= 11 + 18 = 29$ cm (2).	1+1+1+2

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** are equal and bisect each other	Property of a rectangle.
2	**(b)** 33°	A diagonal of a rhombus bisects the angle at the vertex.
3	**(c)** 29 cm	Each new side is half of a side of the original triangle.
4	**(d)** 128 cm^2	Area $= \frac{1}{2}d^2$.
5	**(c)** 144°	One part $= 360^\circ \div 15 = 24^\circ$.
6	**(a)** 22	Opposite angles are equal.
7	**(d)** 60°	Adjacent angles add up to 180°.
8	**(b)** 34 cm	Diagonals bisect each other, so $AC = 2 \times OA$.
9	**(a)** 360°	Split into two triangles.
10	**(d)** 28°	A diagonal of a rhombus bisects the angle at the vertex.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$EF = 13$ cm, $FD = 24$ cm, $DE = 18$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 13 + 24 + 18 = 55$ cm (1).	1+1
12	$\angle C = \angle A = 60^\circ$ (opposite angles); $\angle B = 180^\circ - 60^\circ = 120^\circ$ (adjacent angles) (1); $\angle D = \angle B = 120^\circ$ (1).	1+1
13	Diagonal $= \sqrt{5^2 + 12^2}$; $= \sqrt{169} = 13$ cm (1); Perimeter $= 2(5 + 12) = 34$ cm (1).	1+1
14	$EF = 12$ cm, $FD = 10$ cm, $DE = 21$ cm (midpoint theorem: each side is half the parallel side) (1); Perimeter $= 12 + 10 + 21 = 43$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Opposite angles are equal; $2x + 10 = x + 27$ gives $x = 17$ (1); Angles $A = C = 44^\circ$ (1); $B = D = 180^\circ - 44^\circ = 136^\circ$ (1).	1+1+1
16	Adjacent angles add up to 180°: $x + 3x = 180^\circ$ (1); $x = 45^\circ$ and the other is 135° (1); Opposite angles are equal: the four angles are $45^\circ, 135^\circ, 45^\circ, 135^\circ$ (1).	1+1+1
17	Diagonals bisect each other, so $OA = OC$; $4x + 9 = 2x + 23$ gives $x = 7$ (1); $OA = 37$ cm (1); $AC = 2 \times 37 = 74$ cm (1).	1+1+1
	OR: Opposite angles are equal; $4x + 7 = x + 76$ gives $x = 23$ (1); Angles $A = C = 99^\circ$ (1); $B = D = 180^\circ - 99^\circ = 81^\circ$ (1).	1+1+1
18	Adjacent angles add up to 180°: $x + 2x = 180^\circ$ (1); $x = 60^\circ$ and the other is 120° (1); Opposite angles are equal: the four angles are $60^\circ, 120^\circ, 60^\circ, 120^\circ$ (1).	1+1+1
	OR: Diagonals bisect each other, so $OA = OC$; $3x + 8 = 4x + 6$ gives $x = 2$ (1); $OA = 14$ cm (1); $AC = 2 \times 14 = 28$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AD = 11$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 12$ cm (1); Hence $AE = EC$ (2).	1+1+1+2
	OR: In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 6$ cm and $QR = PS = 8$ cm (1); Perimeter $= 12 + 16 = 28$ cm (2).	1+1+1+2
20	$AD = 8$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 15$ cm (1); Hence $AE = EC$ (2).	1+1+1+2

	OR: In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 9$ cm and $QR = PS = 14.5$ cm (1); Perimeter $= 18 + 29 = 47$ cm (2).	1+1+1+2

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

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